

Structural Team Discussion

Meeting Summary (10/09/26)

The following key points were discussed with the Structural Team regarding the Steel Beam Estimation automation:

1. Change in Requirement – Beam Dimension Measurement & Extraction

Previous Approach:

Steel Beam Estimation was based on three documents:

- General Notes
- Framing Plan
- Reinforcement Plan

As discussed with the Estimation Team, the **Framing Plan** was used to measure and extract **Beam Length, Width, and Depth**.

Revised Requirement:

The Structural Team has requested that **Beam Length, Width, and Depth** be measured and extracted from the **Reinforcement Plan** instead.

Reason:

According to the Structural Team, the beam length represented in the Reinforcement Plan is expected to provide a **more accurate representation of the actual beam dimensions** compared to the Framing Plan.

My Comment:

We will **try implementing this approach** and evaluate the accuracy and feasibility of extracting the beam dimensions from the Reinforcement Plan.

2. Change in Objective – Drawing Generation

Present Objective:

The primary objective is to **estimate quantities from structural drawings**.

Revised Objective Expectation:

The Structural Team asked whether the system could be extended to **generate structural drawings similar to those prepared by Structural Engineers**, using **ETABS drawings and associated structural data** as the input.

My Comment:

This is not currently feasible with the AI model. We have already explored a similar approach during the Slab Concrete estimation, where the model was tested for generating slab polylines. The results showed that the model is not reliable for accurately generating structural drawings.

Therefore, the current scope will remain focused on drawing interpretation and quantity estimation, rather than automated structural drawing generation.